G1 hardware logs: what the data show, and how to reproduce the figures

Right-arm bench test (Port-Hamiltonian torque law vs position-PD) and one grasp run of the Unitree G1, recorded on 18 September 2026

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1. What we learned

The four CSV logs Ammar sent (30 September) are the same runs that Section V of the thesis draft reports. Read carefully, they tell a different and more useful story than the draft does.

- **The arm accepts torque commands.** This was the open question the bench test was built for, and the answer is yes. The thesis does not say it yet; it should, because it is the solid positive result.
- **“PH tracks worse” is the wrong reading.** Both control laws leave the arm sagging below its target. The gravity model predicts about 3.0 Nm at the shoulder for this pose; the arm actually needs 5.7 to 7.5 Nm. The PH law is softer (18 against 60 Nm/rad), so the same missing torque makes it sag further. The problem is the gravity model, not the control law.
- **The hold segment is mostly friction.** In both runs the arm settles within about 0.6 s and then sits frozen for the remaining 4.4 s. There is half a second of real behaviour, logged at only 10 Hz.
- **The waist is not fixed.** It leans forward by up to 0.12 rad (about 7°) whenever the arm reaches out, although its command never changes. Proposition 1 assumes it is fixed.
- **The large error at the end of the grasp run is not a fault.** In the last segment (side down) the arm is handed back to Unitree's own controller, so the logged target no longer applies.
- **The hand works.** The fingers close, settle at about 2 to 2.5 N each and hold through the lift. The thumb-yaw sensor is faulty (known), and the ring sensor reads about −0.7 N with nothing touching it.

> **Bottom line.** PH control looks usable on the real arm. What the bench test measured is mainly the error of the gravity model. Fix the model, then test what PH control is actually for: safe response to contact (push the arm and watch it come back), not free-space accuracy, where a stiff PD wins by design.

2. Setting up and running (Windows PowerShell)

Everything lives in one folder, **G1_hardware_analysis**, which goes into Documents. The script finds its data and writes its figures relative to its own location, so the folder can be moved anywhere without editing the code.

Item	What it is
make_figs.py	The analysis script. Reads data\, writes figures\.
run_figures.ps1	One-click runner: sets up Python on the first run, then calls make_figs.py.
requirements.txt	The three libraries needed: numpy, pandas, matplotlib.
data\	The four robot logs (CSV) and Ammar's REAL_LOGS_ARCHITECTURE.md, which documents every column.
figures\	Output: fig1 to fig7 as PNG (300 dpi, for slides) and PDF (vector, for LaTeX), and results_summary.csv.
report\	This report.
README.md	Short version of this section.

Step 1. Put the folder in Documents (once)

Save G1_hardware_analysis.zip in Downloads, open PowerShell and run:

```
$docs = [Environment]::GetFolderPath("MyDocuments")   # correct even if OneDrive moved Documents
Expand-Archive -Path "$HOME\Downloads\G1_hardware_analysis.zip" -DestinationPath $docs
cd "$docs\G1_hardware_analysis"
```

Step 2. Make the figures

The easy way. The first run takes a minute: it creates a private Python environment in a .venv subfolder and installs the three libraries there, so nothing else on the computer changes. Later runs take a few seconds. When it finishes, the figures folder opens in Explorer.

```
powershell -ExecutionPolicy Bypass -File .\run_figures.ps1
```

If Python with numpy, pandas and matplotlib is already installed, calling the script directly also works:

```
python make_figs.py
```

Step 3 (for Ammar). Use the logs where they already are

Ammar does not need to copy anything. The --data option points the script at his existing log folder, and the timestamp options choose particular runs, for example one of the four other grasp runs in Successfull_grasps. Without options the newest file of each kind is used.

```
python make_figs.py --data "C:\path\to\Real_logs"
python make_figs.py --data "C:\path\to\Successfull_grasps" --grasp-run 20260918_185109
python make_figs.py --bench-ph 20260918_183908 --bench-pd 20260918_184048
python make_figs.py --out "C:\path\to\thesis\figures"
```

The same options work after run_figures.ps1. Note that --data changes the folder for all four logs, so the bench-test files must be in that folder too (or copy the grasp-run files into data\ instead).

If something goes wrong

Message	Cause and fix
Python 3 not found	Install Python 3.9 or newer from python.org and tick “Add python.exe to PATH”. If typing python opens the Microsoft Store, install from python.org instead.
…cannot be loaded because running scripts is disabled	Start the runner exactly as shown, with -ExecutionPolicy Bypass. This affects only that one call.
[ERROR] no arm_ph_ph_*.csv in …	The data folder is empty or wrong. Check the path, or pass --data.
[ERROR] not found: …_.csv	No log with that timestamp. Check the file names in the data folder.
pip cannot download	No internet or a proxy. Install the three libraries by hand, then run python make_figs.py.

3. How the script works

make_figs.py runs top to bottom in about five seconds. It has four parts: finding the files, shared settings and helpers, one block per figure, and the summary table at the end.

3.1 The input files

The logs come from two different programs on the robot, which is easy to miss. The first two are the arm bench test; the last two are one grasp run, logged at the same moment by grasp_ph.py. All are sampled at 10 Hz.

File	Produced by	Contents
arm_ph_ph_<time>.csv	g1_hw_arm_ph_control.py --mode ph	Bench test, PH torque law in the hold. Joint angles, targets, errors, commanded torque and the model gravity torque g(q) for the 7 right-arm joints.
arm_ph_position_<time>.csv	g1_hw_arm_ph_control.py --mode position	Same choreography with plain position-PD (kp 60, kd 1.5). Torque columns are zero; g(q) is not computed.
arm_positions_<time>.csv	grasp_ph.py (arm log)	Grasp run, arm under position-PD with IK targets: all 29 joint angles plus targets and errors for the 17 arm and waist joints.
hand_state_<time>.csv	grasp_ph.py (hand log)	Grasp run, both hands under the PH admittance law: angle, force (grams), admittance reference and force-mode target for 12 finger joints.

3.2 Finding the files

find_log() looks in the data folder for a file that starts with the right prefix. With a timestamp it takes that exact file; without one it takes the newest. The hand log is matched to the arm log of the same grasp run by timestamp, so the two always belong together. clock() turns the timestamp in the file name into a readable date for the figure subtitles.

```
F_PH   = find_log("arm_ph_ph_",       args.bench_ph)
F_PD   = find_log("arm_ph_position_", args.bench_pd)
F_ARM  = find_log("arm_positions_",   args.grasp_run)
F_HAND = find_log("hand_state_",      <same timestamp as F_ARM>)
```

3.3 Settings and helpers

Name	Meaning
JOINTS	The three joints plotted: shoulder pitch, elbow, wrist pitch, with their PH stiffness Kp (15, 10, 3 Nm/rad) from g1_hw_arm_ph_control.py. The other four right-arm joints move less than 0.02 rad.
PD_KP = 60	Stiffness of the position-PD baseline (the firmware's kp).
SAFETY_KP = 3	Stiffness of the weak firmware PD “floor” that runs underneath the PH torque law.
steady(d)	The hold segment minus its first second, so the switch-over transient is excluded and only the settled state is averaged.
suptitle(), shade(), save()	Layout: title and subtitle above the plot, grey bands for segments, and saving each figure as PNG and PDF.
note()	Records every number a figure quotes; all of them are written to results_summary.csv.

3.4 The calculations

Most figures plot logged columns directly. Figures 3 and 4 need a short calculation, and it rests on one physical fact: once the arm stands still, the torque from the controller must exactly balance whatever holds the arm down (gravity, plus friction and any model error). The controller's torque can be read off the log, so it measures how much torque the pose really needs.

With $e = q - q_{des}$ the settled error (radians) and joint velocity zero:

```
Position-PD run   firmware torque = kp (q_des - q) = -60 e
                  -> holding torque  tau_PD = 60 |e_PD|

PH run            firmware torque = tau_cmd + 3 (q_des - q)
                  tau_cmd = g(q) - Kp e - Kd qdot     (logged; qdot = 0 at rest)
                  -> holding torque  tau_PH = |tau_cmd| + 3 |e_PH|
                     split into  g(q)            (model gravity, logged)
                               + |tau_cmd| - g   (impedance feedback)
                               + 3 |e_PH|         (safety floor)

"If tau were ignored"   only the floor would hold the arm:  e = tau_PH / 3
```

In code (Figure 3 block), per joint:

```
g       = abs(sp[j + "_g_Nm"].mean())          # model gravity torque
tau     = sp[j + "_tau_cmd_Nm"].mean()         # torque the PH law sent
e       = sp[j + "_qerr_rad"].mean()           # settled PH error
fb      = abs(tau) - g                         # feedback part of tau
floor   = abs(SAFETY_KP * e)                   # safety-floor part
pd_hold = abs(PD_KP * spd[j + "_qerr_rad"].mean())   # PD run
```

The position-PD run does not log g(q), because that mode never computes it. The script therefore compares both runs against the g(q) logged in the PH run. The two settled poses differ by at most 0.1 rad (elbow), which changes g(q) by a few percent at most, far less than the factors of 2 to 5 at stake.

3.5 Output

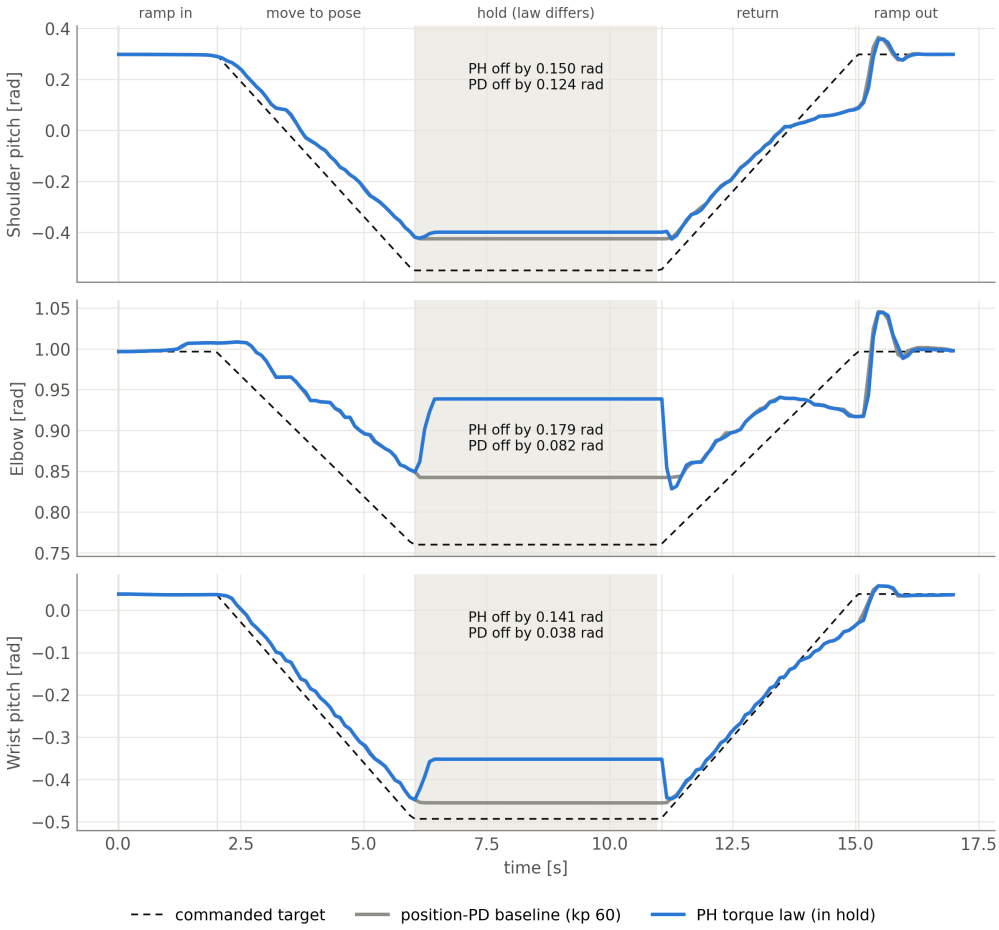
Seven figures, each as PNG and PDF, plus results_summary.csv with every quoted number (listed in the appendix). Subtitles are computed from the data, so they stay correct when the script runs on another grasp run.

4. The figures, one by one

Figure 1.

Arm bench test: neither law reaches the target pose

Right arm, same choreography run twice (PH 18 Sep 2026, 18:39; PD 18 Sep 2026, 18:40). Only the grey hold band uses a different law.



What it shows

Three right-arm joints over the full 17 s bench test. Dashed black is the commanded target, grey the position-PD run, blue the PH run. The two runs use the same law everywhere except in the grey hold band.

How to read it

Outside the band the blue and grey lines lie on top of each other, as they should. Inside it, look at the vertical gap between each line and the dashed target: that is the tracking error the thesis reports.

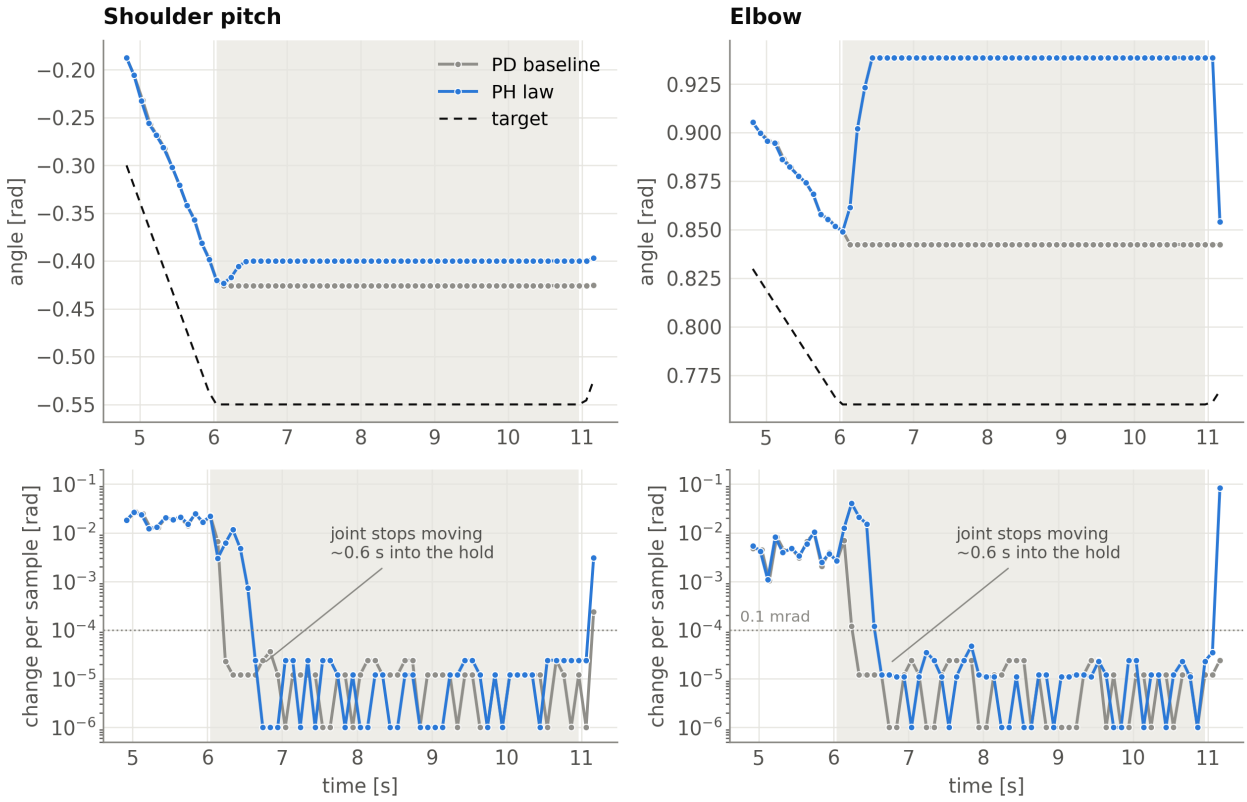
What it means

Both laws stop short of the target in the same direction, which is the direction gravity pulls. The PH run sags further on every joint. The ramp-out bump near 15.5 s is the hand-back to Unitree's controller and appears in both runs.

Figure 2.

The 5 s hold is a 0.6 s transient, then a frozen equilibrium

Every logged sample shown (10 Hz). After the switch, the joints stop within about six samples in both runs: what the hold reports is a static offset held by friction, not tracking.



What it shows

A zoom on the hold for shoulder pitch and elbow, with every logged sample drawn as a dot. The bottom panels show how far the joint moved between two samples, on a log scale.

How to read it

Above the dotted 0.1 mrad line the joint is moving; below it the changes are sensor noise. Both runs drop below the line about six samples into the hold and stay there.

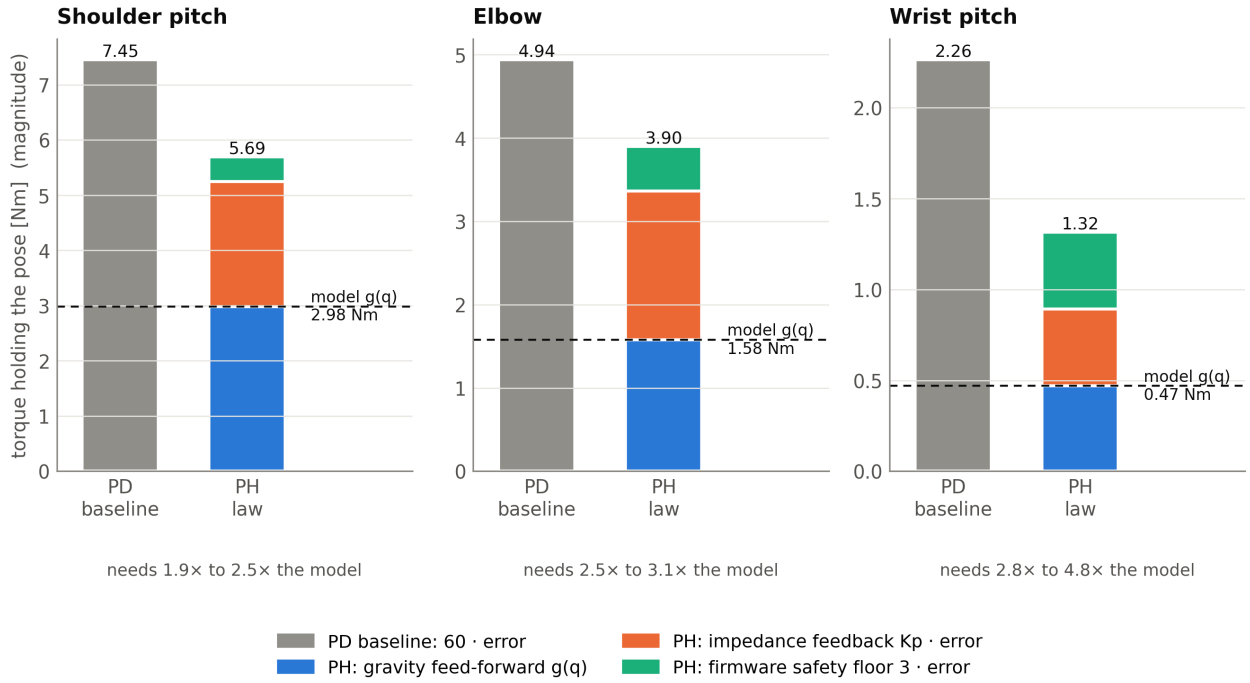
What it means

Averages over the hold describe a joint stuck in place by friction, not a controller tracking a target. Only the first 0.6 s shows the control law acting, and 10 Hz gives just six points of it. Log at 50 Hz next time.

Figure 3.

The gravity model under-predicts the torque the arm needs

Torque that actually held the arm still during the hold, from each run. Both runs, measured independently, need 1.9 to 4.8 times Pinocchio's g(q). The larger PH error is what a lower stiffness (18 vs 60 Nm/rad on shoulder pitch) does with the same missing torque.



What it shows

For each joint, the torque that actually held the arm still (section 3.4). The grey bar is from the position-PD run. The stacked bar is from the PH run: model gravity g(q) in blue, impedance feedback in orange, safety floor in green. The dashed line is g(q).

How to read it

If the gravity model were right, the stacked bar would be almost all blue and the grey bar would reach the dashed line. Instead both bars stand well above it, by the factor printed under each panel.

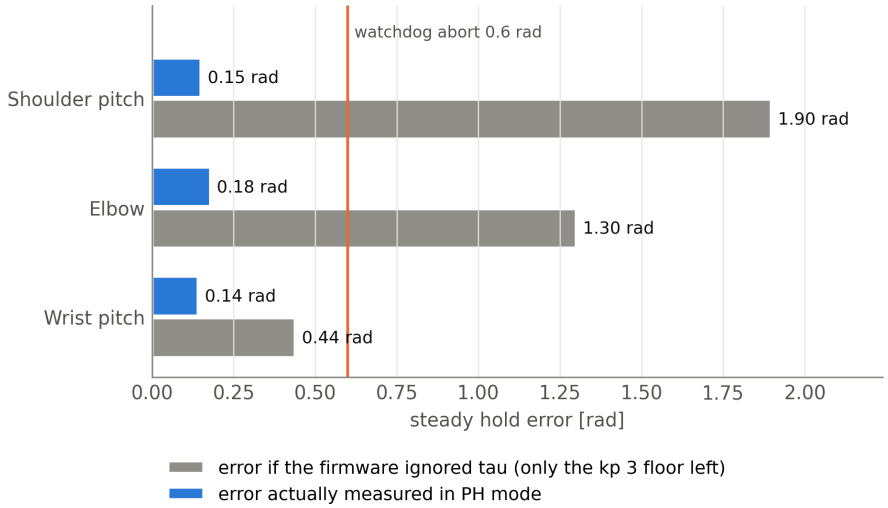
What it means

Two independent runs agree that the pose needs 1.9 to 4.8 times the modelled gravity torque. Likely causes: missing or wrong mass in the URDF (the hand), joint friction, or the kp and tau fields not being in the SI units assumed. The PH error is larger simply because its stiffness is lower (18 against 60 Nm/rad at the shoulder). The draft's percentages (75 %, 89 %) also leave out the green safety-floor part.

Figure 4.

The arm firmware does apply the torque command

Without tau the weak floor would let shoulder and elbow sag past the watchdog.
The measured error is 3 to 13 times smaller, so tau reaches the motors.



What it shows

For each joint, the settled PH error (blue) next to the error the arm would show if the firmware ignored tau and only the weak kp 3 floor held it (grey). The orange line is the 0.6 rad watchdog that aborts the run.

How to read it

Grey past orange means that without tau the run would have been aborted. It was not aborted, and the blue errors are 3 to 13 times smaller than grey.

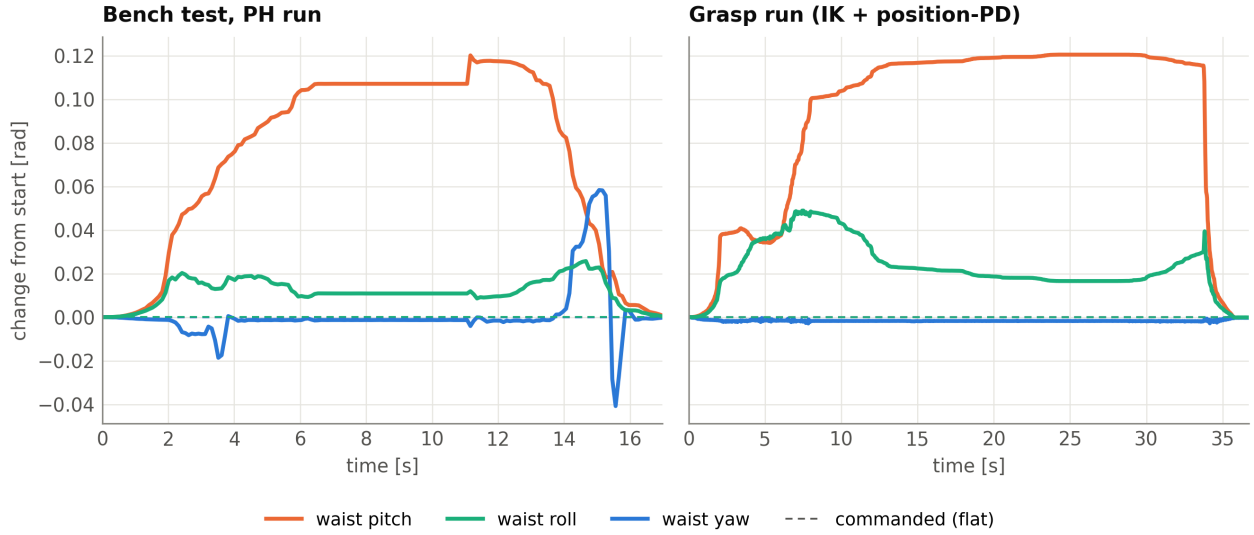
What it means

The commanded torque reaches the motors. This is the answer to the question the bench test was built for, and it belongs in the thesis as a result.

Figure 5.

The waist is not static

Waist pitch leans up to 0.12 rad forward while the arm reaches out, although its command never changes. Proposition 1 assumes these joints are fixed, and g(q) depends on them.



What it shows

Change of the three waist angles from their start value, in the PH bench run (left) and the grasp run (right). The dashed lines are the commands, which stay flat.

How to read it

Any gap between a solid line and zero is waist motion that nobody commanded.

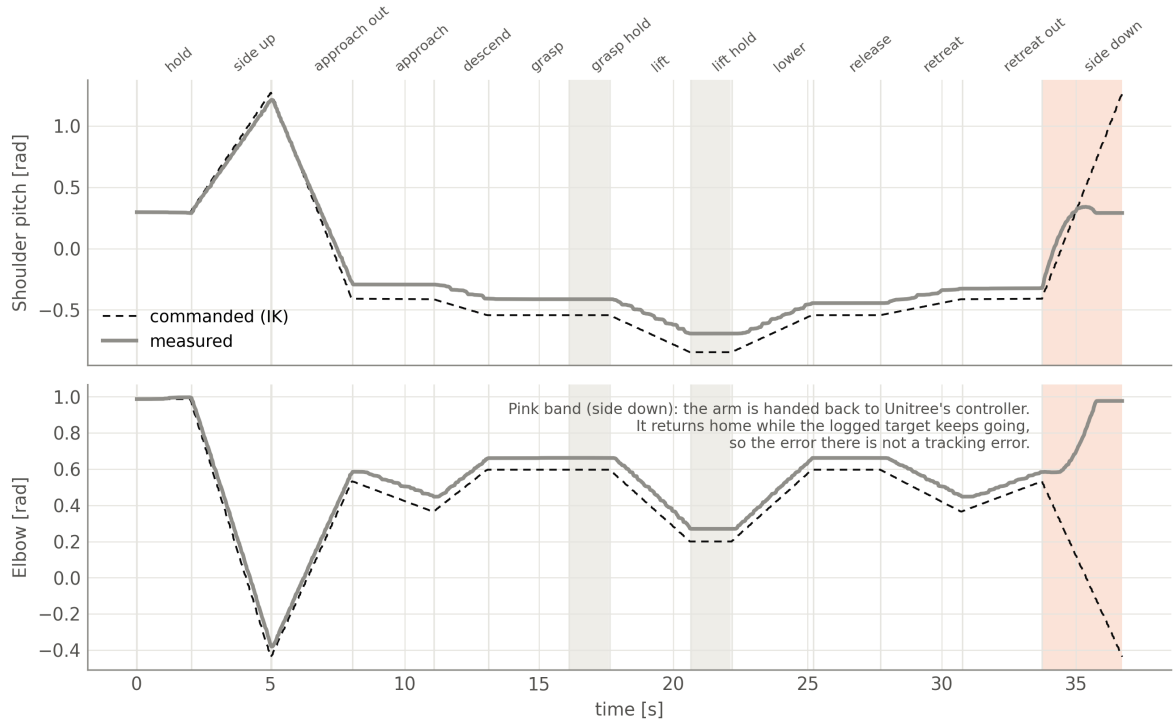
What it means

Waist pitch leans forward by up to 0.12 rad when the arm reaches out, and roll moves up to 0.05 rad. The position-PD holding the waist is not stiff enough against the arm's reaction. Section II-C says the waist is effectively static and Proposition 1 assumes fixed joints; both need a sentence. The gravity term in the bench script already uses the measured waist angle, which is right.

Figure 6.

Grasp run: a steady 0.13–0.17 rad shoulder offset, and a false alarm at the end

Same sag as in the bench test. Excluding side down, the largest shoulder-pitch error is 0.17 rad (it reaches 0.98 rad inside side down).



What it shows

Shoulder pitch and elbow through the 14 segments of the grasp run: IK target dashed, measured in grey. Grey bands mark grasp hold and lift hold; the pink band marks side down.

How to read it

Look at the constant gap between grey and dashed from approach to retreat, then at the pink band.

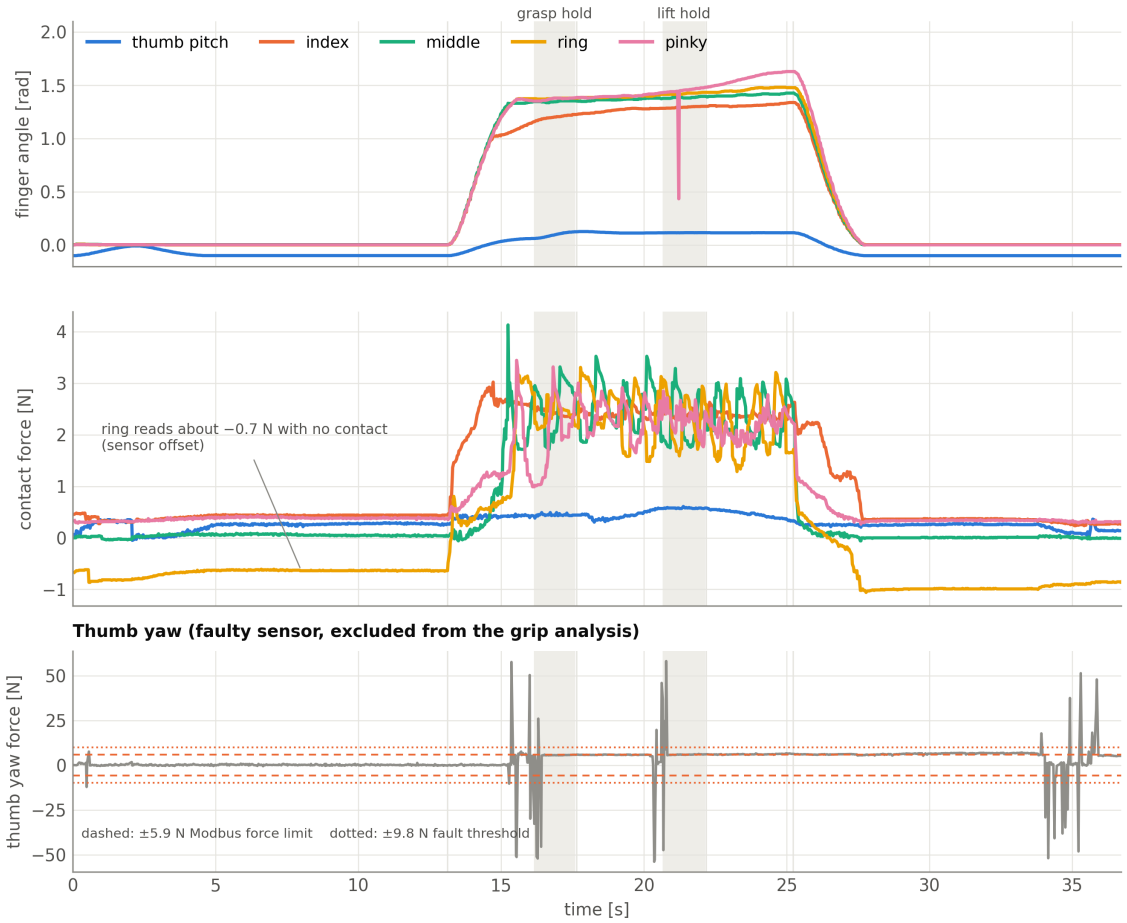
What it means

The same gravity sag as in the bench test shows up here: 0.13 rad at the shoulder in grasp hold, up to 0.17 rad in lift. At the hand this is probably several centimetres, which matters for where the fingers land. In side down the error reaches 0.98 rad, but there the arm is being handed back to Unitree's controller and returns home while the logged target keeps moving. It is not a tracking error and should be left out; this is likely the unexplained error mentioned in Section V.

Figure 7.

Hand: the grip is set in grasp hold and survives the lift

Right hand, grasp run of 18 Sep 2026, 18:51. The four fingers hold 2.1–2.4 N on average from grasp hold through lift hold. Logged force-mode target τd = 0.75 Nm on every right-hand joint; the thesis text says 2 Nm (1 Nm thumb yaw).



What it shows

Right hand during the grasp run: finger angles (top), measured contact force (middle) and the thumb-yaw force (bottom). Grey bands mark grasp hold and lift hold.

How to read it

In the middle panel, compare the force level inside the two grey bands. If the object slipped during the lift, the forces would drop in the second band.

What it means

Index, middle, ring and pinky hold about 2 to 2.5 N from grasp hold through lift hold, so the grip carries the object. Two sensor issues: thumb yaw spikes to about 58 N and sits on the 5.9 N limit (known fault), and the ring finger reads about −0.7 N with no contact, so its force is uncertain by that much. One point to check: the log records a force-mode target of 0.75 Nm on every right-hand joint, while the thesis text says 2 Nm (1 Nm on thumb yaw).

5. Assumptions and limits

- All logs are sampled at 10 Hz, while the controller runs at 50 Hz. Fast behaviour is under-sampled.
- Joint velocity and the motors' own torque estimate are not logged, so torques are inferred from errors and gains (section 3.4), assuming the joints are at rest.

- The kp, kd and tau fields of Unitree's arm interface are taken as Nm/rad, Nm·s/rad and Nm. The thesis already notes they are not independently calibrated; a units mismatch would change the factors in Figure 3.
- Friction means a held joint can rest anywhere inside a band. Both bench runs approached the pose from the same side, so part of the missing torque in Figure 3 may be friction rather than gravity.
- The zip holds one grasp run. Table VI of the thesis averages five; the other four can be run through the same script with --grasp-run.

6. Next steps

For the thesis

- State the positive result: the G1 arm accepts and applies torque commands (Figure 4).
- Reframe Section V-A: the hold error measures gravity-model error divided by stiffness, not PH against PD (Figure 3).
- Qualify the fixed-waist assumption (Figure 5) and explain the side-down error (Figure 6).
- Reconcile the 0.75 Nm logged grip target with the 2 Nm in the text.

For the next hardware session

- Log at 50 Hz, and add joint velocity (dq), the motors' torque estimate (tau_est) and the arm_sdk hand-over weight.
- Calibrate gravity: hold four or five poses, read tau_est, compare with g(q), and check the hand mass in the URDF.
- Measure friction: approach the hold pose once from above and once from below.
- Compare fairly: run PH with stiffness matched to the baseline (Kp + 3 = 60 on shoulder and elbow).
- Test what PH is for: push the arm during the hold and release it. Compare PH and PD on contact force, return to pose and energy decay.

Appendix. Numbers behind the figures

The content of figures\results_summary.csv for the runs in this package. Holding torques are magnitudes.

Figure	Quantity	Value	Unit
fig1	Shoulder pitch: steady hold error, PH	0.150	rad
fig1	Shoulder pitch: steady hold error, PD	0.124	rad
fig1	Elbow: steady hold error, PH	0.179	rad
fig1	Elbow: steady hold error, PD	0.082	rad
fig1	Wrist pitch: steady hold error, PH	0.141	rad
fig1	Wrist pitch: steady hold error, PD	0.038	rad
fig3	Shoulder pitch: model gravity torque g(q)	2.981	Nm
fig3	Shoulder pitch: holding torque, PH run (tau + floor)	5.693	Nm
fig3	Shoulder pitch: holding torque, PD run (60 x error)	7.454	Nm
fig3	Elbow: model gravity torque g(q)	1.577	Nm
fig3	Elbow: holding torque, PH run (tau + floor)	3.900	Nm
fig3	Elbow: holding torque, PD run (60 x error)	4.941	Nm
fig3	Wrist pitch: model gravity torque g(q)	0.470	Nm
fig3	Wrist pitch: holding torque, PH run (tau + floor)	1.315	Nm
fig3	Wrist pitch: holding torque, PD run (60 x error)	2.265	Nm
fig4	Shoulder pitch: PH error measured	0.150	rad
fig4	Shoulder pitch: PH error if tau were ignored	1.898	rad
fig4	Elbow: PH error measured	0.179	rad
fig4	Elbow: PH error if tau were ignored	1.300	rad
fig4	Wrist pitch: PH error measured	0.141	rad
fig4	Wrist pitch: PH error if tau were ignored	0.438	rad
fig5	largest forward lean of waist pitch	0.120	rad
fig6	shoulder pitch error in grasp hold (median)	0.130	rad
fig6	largest shoulder pitch error, side_down excluded	0.171	rad
fig6	largest shoulder pitch error inside side_down	0.981	rad
fig7	thumb pitch: mean force in grasp hold	0.444	N
fig7	thumb pitch: mean force in lift hold	0.577	N
fig7	index: mean force in grasp hold	2.428	N
fig7	index: mean force in lift hold	2.386	N
fig7	middle: mean force in grasp hold	2.575	N
fig7	middle: mean force in lift hold	2.285	N
fig7	ring: mean force in grasp hold	2.421	N
fig7	ring: mean force in lift hold	2.337	N
fig7	pinky: mean force in grasp hold	1.937	N
fig7	pinky: mean force in lift hold	2.304	N
fig7	logged force-mode target tau_d	0.750	Nm
fig7	thumb yaw peak |force|	58.085	N